Contents

1 Course Supplement: Practice Questions & Solutions

1.0.1 Ordinary Differential Equations (ODEs) & Linear Algebra

This document serves as a supplementary guide to the main course PDF, capturing all additional homework and practice questions from the handwritten notes and lecture materials.

1.1 Module 2: Solving First-Order ODEs

1.1.1 2.1 Variable Separable Method

Question 1: Solve: $y(1+x^2)^{1/2}dy + x\sqrt{1+y^2}dx = 0$

Solution:

1. **Separate Variables:** Move the dx term to the right side:
$$y(1+x^2)^{1/2}dy = -x\sqrt{1+y^2}dx$$
Divide by $(1+x^2)^{1/2}$ and $\sqrt{1+y^2}$:
$$\frac{y}{\sqrt{1+y^2}}dy = -\frac{x}{\sqrt{1+x^2}}dx$$

2. **Integrate both sides:**
$$\int \frac{y}{\sqrt{1+y^2}}dy = -\int \frac{x}{\sqrt{1+x^2}}dx$$
(Multiply and divide by 2 on both sides to match the exact derivative of the inside of the root)
$$\frac{1}{2}\int (1+y^2)^{-1/2}(2y)dy = -\frac{1}{2}\int (1+x^2)^{-1/2}(2x)dx$$
$$\frac{1}{2}\left[\frac{(1+y^2)^{1/2}}{1/2}\right] = -\frac{1}{2}\left[\frac{(1+x^2)^{1/2}}{1/2}\right] + C$$
$$\sqrt{1+y^2} = -\sqrt{1+x^2} + C$$

3. **Final General Solution:**
$$\sqrt{1+x^2} + \sqrt{1+y^2} = C$$

Question 2: Solve: $\frac{dy}{dx} = (4x+y+1)^2$

Solution:

1. **Substitution:** This equation is not separable in its current form, but it can be reduced to separable. Let $v = 4x+y+1$. Differentiate with respect to x:
$$\frac{dv}{dx} = 4 + \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{dv}{dx} - 4$$

2. **Substitute into the equation:**
$$\frac{dv}{dx} - 4 = v^2 \implies \frac{dv}{dx} = v^2 + 4$$

3. **Separate Variables and Integrate:**

$$\frac{dv}{v^2+4} = dx$$

$$\int \frac{1}{v^2+2^2} dv = \int dx$$

$$\frac{1}{2}\tan^{-1}\left(\frac{v}{2}\right) = x + C_1$$

4. **Resubstitute v and simplify:**

$$\tan^{-1}\left(\frac{4x+y+1}{2}\right) = 2x + 2C_1$$

Final Answer:

$$4x + y + 1 = 2\tan(2x + C)$$

(where $C = 2C_1$)

Question 3: Solve: $(x+1)\frac{dy}{dx} + 1 = 2e^{-y}$

Solution:

1. **Rearrange:**

$$(x+1)\frac{dy}{dx} = 2e^{-y} - 1$$

2. **Separate Variables:**

$$\frac{dy}{2e^{-y}-1} = \frac{dx}{x+1}$$

Multiply numerator and denominator of the LHS by e^y:

$$\frac{e^y}{2-e^y} dy = \frac{dx}{x+1}$$

3. **Integrate both sides:**

$$\int \frac{e^y}{2-e^y} dy = \int \frac{1}{x+1} dx$$

Let $u = 2 - e^y \implies du = -e^y dy$:

$$-\int \frac{-e^y}{2-e^y} dy = \ln|x+1| + \ln C$$

$$-\ln|2-e^y| = \ln|x+1| + \ln C$$

4. **Simplify using Log rules:**

$$\ln|2-e^y|^{-1} = \ln|C(x+1)| \implies \frac{1}{2-e^y} = C(x+1)$$

Final Answer:

$$(x+1)(2-e^y) = K$$

(where $K = 1/C$)

1.1.2 2.2 Homogeneous Differential Equations

Question 4: Solve: $\frac{dy}{dx} + \frac{x-2y}{2x-y} = 0$

Solution:

1. **Standard Form:** $\frac{dy}{dx} = \frac{2y-x}{2x-y}$
2. **Substitute:** Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$

$$v + x\frac{dv}{dx} = \frac{2vx - x}{2x - vx} = \frac{x(2v-1)}{x(2-v)} = \frac{2v-1}{2-v}$$

3. **Separate Variables:**

$$x\frac{dv}{dx} = \frac{2v-1}{2-v} - v = \frac{2v-1-2v+v^2}{2-v} = \frac{v^2-1}{2-v}$$

$$\frac{2-v}{v^2-1}dv = \frac{dx}{x}$$

4. **Partial Fractions:** $\frac{2-v}{(v-1)(v+1)} = \frac{A}{v-1} + \frac{B}{v+1}$ $2 - v = A(v+1) + B(v-1) \implies A = 1/2, B = -3/2$

$$\int \left(\frac{1/2}{v-1} - \frac{3/2}{v+1}\right) dv = \int \frac{dx}{x}$$

$$\frac{1}{2}\ln|v-1| - \frac{3}{2}\ln|v+1| = \ln|x| + \ln C$$

Multiply by 2:

$$\ln|v-1| - 3\ln|v+1| = 2\ln|x| + \ln K$$

$$\ln\left|\frac{v-1}{(v+1)^3}\right| = \ln(Kx^2) \implies \frac{v-1}{(v+1)^3} = Kx^2$$

5. **Resubstitute $v = y/x$:**

$$\frac{y/x - 1}{(y/x+1)^3} = Kx^2 \implies \frac{\frac{y-x}{x}}{\frac{(y+x)^3}{x^3}} = Kx^2 \implies \frac{x^2(y-x)}{(y+x)^3} = Kx^2$$

Final Answer:

$$y - x = K(y+x)^3$$

Question 5: Solve: $(x^2 + y^2)dy = xydx$

Solution:

1. **Standard Form:** $\frac{dy}{dx} = \frac{xy}{x^2+y^2}$
2. **Substitute:** Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$

$$v + x\frac{dv}{dx} = \frac{x(vx)}{x^2 + (vx)^2} = \frac{v}{1+v^2}$$

3. **Separate Variables:**

$$x\frac{dv}{dx} = \frac{v}{1+v^2} - v = \frac{v - v - v^3}{1+v^2} = \frac{-v^3}{1+v^2}$$

$$\frac{1+v^2}{v^3}dv = -\frac{dx}{x} \implies \left(v^{-3} + \frac{1}{v}\right) dv = -\frac{dx}{x}$$

4. **Integrate:**

$$-\frac{1}{2v^2} + \ln|v| = -\ln|x| + C$$

5. **Resubstitute $v = y/x$:**

$$-\frac{x^2}{2y^2} + \ln\left|\frac{y}{x}\right| + \ln|x| = C \implies -\frac{x^2}{2y^2} + \ln|y| - \ln|x| + \ln|x| = C$$

Final Answer:

$$\ln|y| - \frac{x^2}{2y^2} = C$$

Question 6: Solve: $(x^3 - 3xy^2)dx + (y^3 - 3x^2y)dy = 0$

Solution:

1. **Standard Form:** $\frac{dy}{dx} = \frac{3xy^2 - x^3}{y^3 - 3x^2y}$
2. **Substitute:** Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$

$$v + x\frac{dv}{dx} = \frac{3v^2 - 1}{v^3 - 3v}$$

3. **Separate Variables:**

$$x\frac{dv}{dx} = \frac{3v^2 - 1}{v^3 - 3v} - v = \frac{3v^2 - 1 - v^4 + 3v^2}{v^3 - 3v} = \frac{-v^4 + 6v^2 - 1}{v^3 - 3v}$$

$$\frac{v^3 - 3v}{v^4 - 6v^2 + 1}dv = -\frac{dx}{x}$$

4. **Integrate:** Let $u = v^4 - 6v^2 + 1 \implies du = (4v^3 - 12v)dv = 4(v^3 - 3v)dv$

$$\frac{1}{4}\int \frac{du}{u} = -\int \frac{dx}{x}$$

$$\frac{1}{4}\ln|v^4 - 6v^2 + 1| = -\ln x + \ln C \implies \ln(v^4 - 6v^2 + 1) = \ln(C^4x^{-4})$$

5. **Resubstitute $v = y/x$:**

$$\left(\frac{y}{x}\right)^4 - 6\left(\frac{y}{x}\right)^2 + 1 = \frac{K}{x^4}$$

Multiply by x^4: **Final Answer:**

$$y^4 - 6x^2y^2 + x^4 = K$$

Question 7: Solve: $\frac{dy}{dx} = \frac{3xy + y^2}{3x^2}$

Solution:

1. **Substitute:** Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$

$$v + x\frac{dv}{dx} = \frac{3x(vx) + (vx)^2}{3x^2} = \frac{3v + v^2}{3} = v + \frac{v^2}{3}$$

2. **Separate Variables:**
$$x\frac{dv}{dx} = \frac{v^2}{3} \implies \frac{3}{v^2}dv = \frac{dx}{x}$$
3. **Integrate:**
$$3\left(-\frac{1}{v}\right) = \ln|x| + C$$
4. **Resubstitute $v = y/x$:**
$$-\frac{3x}{y} = \ln|x| + C \implies -3x = y\ln|x| + Cy$$

Final Answer:
$$3x + y\ln x + Cy = 0$$

Question 8: Solve: $[x\cos(\frac{y}{x}) + y\sin(\frac{y}{x})]y - [y\sin(\frac{y}{x}) - x\cos(\frac{y}{x})]x\frac{dy}{dx} = 0$

Solution:

1. **Standard Form:**
$$\frac{dy}{dx} = \frac{y[x\cos(y/x) + y\sin(y/x)]}{x[y\sin(y/x) - x\cos(y/x)]}$$
2. **Substitute:** Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$
$$v + x\frac{dv}{dx} = \frac{vx[x\cos v + vx\sin v]}{x[vx\sin v - x\cos v]} = \frac{v(\cos v + v\sin v)}{v\sin v - \cos v}$$
3. **Separate Variables:**
$$x\frac{dv}{dx} = \frac{v\cos v + v^2\sin v - v^2\sin v + v\cos v}{v\sin v - \cos v} = \frac{2v\cos v}{v\sin v - \cos v}$$
$$\frac{v\sin v - \cos v}{v\cos v}dv = \frac{2dx}{x} \implies \left(\tan v - \frac{1}{v}\right)dv = \frac{2dx}{x}$$
4. **Integrate:**
$$\int \tan v dv - \int \frac{1}{v}dv = 2\int \frac{dx}{x}$$
$$\ln|\sec v| - \ln|v| = 2\ln|x| + \ln C \implies -\ln|\cos v| - \ln|v| = \ln(Cx^2)$$
$$\ln\left(\frac{1}{v\cos v}\right) = \ln(Cx^2) \implies \frac{1}{v\cos v} = Cx^2$$
5. **Resubstitute $v = y/x$:**
$$\frac{x}{y\cos(y/x)} = Cx^2 \implies \frac{1}{xy\cos(y/x)} = C$$

Final Answer:
$$xy\cos(y/x) = K$$

Question 9: Solve: $x\frac{dy}{dx} = y(\log y - \log x + 1)$

Solution:

1. **Standard Form:** $\frac{dy}{dx} = \frac{y}{x}\left(\ln\left(\frac{y}{x}\right) + 1\right)$
2. **Substitute:** Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$

$$v + x\frac{dv}{dx} = v(\ln v + 1) = v\ln v + v$$

3. **Separate Variables:**

$$x\frac{dv}{dx} = v\ln v \implies \frac{dv}{v\ln v} = \frac{dx}{x}$$

4. **Integrate:** Let $u = \ln v \implies du = \frac{1}{v}dv$

$$\int \frac{1}{u}du = \int \frac{dx}{x} \implies \ln|u| = \ln|x| + \ln C$$

$$\ln|\ln v| = \ln|Cx| \implies \ln v = Cx$$

5. **Resubstitute** $v = y/x$**: Final Answer:**

$$\ln\left(\frac{y}{x}\right) = Cx \implies y = xe^{Cx}$$

1.1.3 2.3 Linear Differential Equations

Question 10: Solve: $(2y - 3x)dx + xdy = 0$

Solution:

1. **Standard Form:** Divide by xdx:

$$\frac{dy}{dx} + \frac{2}{x}y = 3$$

2. **Identify** $P(x)$ **and find I.F.:** $P(x) = \frac{2}{x}$

$$I.F. = e^{\int \frac{2}{x}dx} = e^{2\ln x} = x^2$$

3. **General Solution Formula:**

$$y \cdot (x^2) = \int 3 \cdot x^2 dx + C$$

$$yx^2 = x^3 + C$$

Final Answer:

$$xy = \frac{x^2}{y} \ldots \text{(Simplified to:)} x^2y - x^3 = C$$

Question 11: Solve: $\frac{dy}{dx} + y\cot x = \cos x$

Solution:

1. **Identify** $P(x)$ **and find I.F.:** $P(x) = \cot x$

$$I.F. = e^{\int \cot x dx} = e^{\ln(\sin x)} = \sin x$$

2. **General Solution Formula:**

$$y \cdot (\sin x) = \int \cos x \cdot \sin x dx + C$$

$$y \sin x = \int \frac{\sin 2x}{2} dx + C \quad \text{OR} \quad y \sin x = \frac{\sin^2 x}{2} + C$$

Final Answer:

$$y \sin x = \frac{\sin^2 x}{2} + C$$

Question 12: Solve: $\frac{dy}{dx} + 2xy = 2e^{-x^2}$

Solution:

1. **Identify $P(x)$ and find I.F.:** $P(x) = 2x$

$$I.F. = e^{\int 2x dx} = e^{x^2}$$

2. **General Solution Formula:**

$$y \cdot (e^{x^2}) = \int 2e^{-x^2} \cdot e^{x^2} dx + C$$

$$ye^{x^2} = \int 2dx + C$$

$$ye^{x^2} = 2x + C$$

Final Answer:

$$y = (2x + C)e^{-x^2}$$

Question 13: Solve: $\frac{dy}{dx} + y \sec x = \tan x$

Solution:

1. **Identify $P(x)$ and find I.F.:** $P(x) = \sec x$

$$I.F. = e^{\int \sec x dx} = e^{\ln(\sec x + \tan x)} = \sec x + \tan x$$

2. **General Solution Formula:**

$$y(\sec x + \tan x) = \int \tan x(\sec x + \tan x) dx + C$$

$$y(\sec x + \tan x) = \int (\sec x \tan x + \sec^2 x - 1) dx + C$$

Final Answer:

$$y(\sec x + \tan x) = \sec x + \tan x - x + C$$

Question 14: Solve: $\cos^2 x \frac{dy}{dx} + y = \tan x$

Solution:

1. **Standard Form:** Divide by $\cos^2 x$:

$$\frac{dy}{dx} + \sec^2 x \cdot y = \tan x \sec^2 x$$

2. **Identify $P(x)$ and find I.F.:** $P(x) = \sec^2 x$

$$I.F. = e^{\int \sec^2 x dx} = e^{\tan x}$$

3. **General Solution Formula:**

$$y \cdot e^{\tan x} = \int e^{\tan x} \tan x \sec^2 x dx + C$$

Let $u = \tan x, du = \sec^2 x dx$:

$$\int u e^u du = u e^u - e^u = e^{\tan x}(\tan x - 1)$$

Final Answer:

$$ye^{\tan x} = e^{\tan x}(\tan x - 1) + C \implies y = \tan x - 1 + Ce^{-\tan x}$$

1.1.4 2.4 Bernoulli's Equations

Question 15: Solve: $x\frac{dy}{dx} = \frac{y^2-1}{y}$

Solution:

1. **Standard Form:**

$$x\frac{dy}{dx} = y - y^{-1} \implies \frac{dy}{dx} - \frac{1}{x}y = -\frac{1}{x}y^{-1}$$

(This is Bernoulli with $n = -1$).

2. **Divide by y^n:** Multiply by y:

$$y\frac{dy}{dx} - \frac{1}{x}y^2 = -\frac{1}{x}$$

3. **Substitution:** Let $z = y^2 \implies \frac{dz}{dx} = 2y\frac{dy}{dx} \implies \frac{1}{2}\frac{dz}{dx} = y\frac{dy}{dx}$.

$$\frac{1}{2}\frac{dz}{dx} - \frac{1}{x}z = -\frac{1}{x} \implies \frac{dz}{dx} - \frac{2}{x}z = -\frac{2}{x}$$

4. **Solve Linear DE for z:**

$$I.F. = e^{\int -\frac{2}{x}dx} = e^{-2\ln x} = x^{-2}$$

$$z \cdot x^{-2} = \int -\frac{2}{x} \cdot x^{-2} dx + C = \int -2x^{-3} dx + C$$

$$zx^{-2} = x^{-2} + C \implies z = 1 + Cx^2$$

5. **Resubstitute $z = y^2$: Final Answer:**

$$y^2 = Cx^2 + 1$$

Question 16: Solve: $\frac{dy}{dx} + y = xy^3$

Solution:

1. **Divide by y^3:**
$$y^{-3}\frac{dy}{dx} + y^{-2} = x$$
2. **Substitution:** Let $z = y^{-2} \implies \frac{dz}{dx} = -2y^{-3}\frac{dy}{dx} \implies -\frac{1}{2}\frac{dz}{dx} = y^{-3}\frac{dy}{dx}$
$$-\frac{1}{2}\frac{dz}{dx} + z = x \implies \frac{dz}{dx} - 2z = -2x$$
3. **Solve Linear DE for z:**
$$I.F. = e^{\int -2dx} = e^{-2x}$$
$$ze^{-2x} = \int -2xe^{-2x}dx + C$$
Integration by parts:
$$-2\left[x\frac{e^{-2x}}{-2} - \int \frac{e^{-2x}}{-2}dx\right] = xe^{-2x} + \frac{1}{2}e^{-2x} + C$$
$$z = x + \frac{1}{2} + Ce^{2x}$$
4. **Resubstitute $z = y^{-2}$: Final Answer:**
$$\frac{1}{y^2} = x + \frac{1}{2} + Ce^{2x}$$

Question 17: Solve: $\frac{dy}{dx} + (\frac{x}{1-x^2})y = x\sqrt{y}$

Solution:

1. **Divide by $y^{1/2}$:**
$$y^{-1/2}\frac{dy}{dx} + \left(\frac{x}{1-x^2}\right)y^{1/2} = x$$
2. **Substitution:** Let $z = y^{1/2} \implies \frac{dz}{dx} = \frac{1}{2}y^{-1/2}\frac{dy}{dx} \implies 2\frac{dz}{dx} = y^{-1/2}\frac{dy}{dx}$
$$2\frac{dz}{dx} + \left(\frac{x}{1-x^2}\right)z = x \implies \frac{dz}{dx} + \frac{x}{2(1-x^2)}z = \frac{x}{2}$$
3. **Solve Linear DE for z:**
$$I.F. = e^{\int \frac{x}{2(1-x^2)}dx}$$
Let $u = 1 - x^2 \implies du = -2xdx \implies xdx = -du/2$.
$$I.F. = e^{-\frac{1}{4}\int \frac{du}{u}} = e^{-\frac{1}{4}\ln(1-x^2)} = (1-x^2)^{-1/4}$$
$$z(1-x^2)^{-1/4} = \int \frac{x}{2}(1-x^2)^{-1/4}dx + C$$
Again, using substitution $u = 1 - x^2$:
$$= -\frac{1}{4}\int u^{-1/4}du = -\frac{1}{4}\left[\frac{u^{3/4}}{3/4}\right] = -\frac{1}{3}u^{3/4} = -\frac{1}{3}(1-x^2)^{3/4} + C$$
$$z = -\frac{1}{3}(1-x^2) + C(1-x^2)^{1/4}$$
4. **Resubstitute $z = \sqrt{y}$: Final Answer:**
$$\sqrt{y} = -\frac{1}{3}(1-x^2) + C(1-x^2)^{1/4}$$

1.2 Module 3: Applications of First-Order ODEs

1.2.1 3.1 Orthogonal Trajectories

Question 18: Find the orthogonal trajectories of $16x^2 + y^2 = a$

Solution:

1. **Differentiate:**
$$32x + 2y\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{16x}{y}$$
2. **Apply Orthogonality Condition:** Replace $\frac{dy}{dx}$ with $-\frac{dx}{dy}$
$$-\frac{dx}{dy} = -\frac{16x}{y} \implies \frac{dx}{x} = 16\frac{dy}{y}$$
3. **Integrate:**
$$\ln|x| = 16\ln|y| + \ln C \implies \ln|x| = \ln(Cy^{16})$$
Final Answer:
$$x = Cy^{16} \quad \text{or} \quad y^{16} = Kx$$

Question 19: Find the orthogonal trajectories of $y^2 = cx^3$

Solution:

1. **Differentiate & Eliminate c:**
$$2y\frac{dy}{dx} = 3cx^2$$
From original, $c = y^2/x^3$.
$$2y\frac{dy}{dx} = 3\left(\frac{y^2}{x^3}\right)x^2 = \frac{3y^2}{x} \implies \frac{dy}{dx} = \frac{3y}{2x}$$
2. **Apply Orthogonality Condition:**
$$-\frac{dx}{dy} = \frac{3y}{2x} \implies -2xdx = 3ydy$$
3. **Integrate:**
$$-\int 2xdx = \int 3ydy \implies -x^2 = \frac{3y^2}{2} + C_1$$
Final Answer:
$$2x^2 + 3y^2 = C$$
(A family of ellipses).

1.3 Module 4: Higher-Order Linear ODEs

1.3.1 4.1 Particular Integral: Exponential Case

Question 20: Solve: $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = e^{3x}$

Solution:

1. **Find y_c:** $m^2 - 6m + 9 = 0 \implies (m-3)^2 = 0 \implies m = 3, 3.$

$$y_c = (C_1 + C_2x)e^{3x}$$

2. **Find y_p:**

$$y_p = \frac{1}{(D-3)^2}e^{3x}$$

Sub $D = 3$: Denominator is 0. (Failure). Multiply by x, diff denominator: $y_p = x\frac{1}{2(D-3)}e^{3x}$. Sub $D = 3 \implies 0$ again. Multiply by x again, diff denominator: $y_p = x^2\frac{1}{2}e^{3x}$.

3. **Final General Solution:**

$$y = y_c + y_p = (C_1 + C_2x)e^{3x} + \frac{x^2}{2}e^{3x}$$

1.3.2 4.2 Particular Integral: Algebraic Case

Question 21: Solve: $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = x$

Solution:

1. **Find y_c:** $m^2 + 2m + 1 = 0 \implies (m+1)^2 = 0 \implies m = -1, -1.$

$$y_c = (C_1 + C_2x)e^{-x}$$

2. **Find y_p:**

$$y_p = \frac{1}{1 + 2D + D^2}x = (1 + 2D + D^2)^{-1}x$$

Using binomial expansion $(1 + X)^{-1} \approx 1 - X$:

$$y_p = [1 - (2D + D^2) + \dots]x = (1 - 2D)x$$

$$y_p = x - 2D(x) = x - 2$$

3. **Final General Solution:**

$$y = (C_1 + C_2x)e^{-x} + x - 2$$

1.3.3 4.3 Particular Integral: Trigonometric & Shift Cases

Question 22: Solve: $\frac{d^2y}{dx^2} + 6y = \sin 4x$

Solution:

1. **Find y_c:** $m^2 + 6 = 0 \implies m = \pm i\sqrt{6}$.
$$y_c = C_1 \cos(\sqrt{6}x) + C_2 \sin(\sqrt{6}x)$$
2. **Find y_p:**
$$y_p = \frac{1}{D^2 + 6} \sin 4x$$
Replace D^2 with $-(4^2) = -16$:
$$y_p = \frac{1}{-16 + 6} \sin 4x = -\frac{1}{10} \sin 4x$$
3. **Final General Solution:**
$$y = C_1 \cos(\sqrt{6}x) + C_2 \sin(\sqrt{6}x) - \frac{1}{10} \sin 4x$$

Question 23: Solve: $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 3x = \sin t$

Solution:

1. **Find x_c:** $m^2 + 2m + 3 = 0 \implies m = \frac{-2\pm\sqrt{4-12}}{2} = -1 \pm i\sqrt{2}$.
$$x_c = e^{-t}(C_1 \cos(\sqrt{2}t) + C_2 \sin(\sqrt{2}t))$$
2. **Find x_p:**
$$x_p = \frac{1}{D^2 + 2D + 3} \sin t$$
Replace D^2 with $-(1^2) = -1$:
$$x_p = \frac{1}{-1 + 2D + 3} \sin t = \frac{1}{2D + 2} \sin t$$
Rationalize by multiplying numerator and denominator by $(D - 1)$ (and extract $1/2$):
$$x_p = \frac{1}{2} \left[\frac{D - 1}{D^2 - 1}\right] \sin t$$
Replace $D^2 = -1$ again:
$$x_p = \frac{1}{2} \left[\frac{D - 1}{-1 - 1}\right] \sin t = -\frac{1}{4}(D - 1) \sin t$$
$$x_p = -\frac{1}{4}(\cos t - \sin t)$$
3. **Final General Solution:**
$$x(t) = e^{-t}(C_1 \cos(\sqrt{2}t) + C_2 \sin(\sqrt{2}t)) - \frac{1}{4} \cos t + \frac{1}{4} \sin t$$

Question 24: Solve: $\frac{d^2y}{dx^2} + 4y = x\sin 2x$

Solution:

1. **Find y_c:** $m^2 + 4 = 0 \implies m = \pm 2i$.

$$y_c = C_1 \cos 2x + C_2 \sin 2x$$

2. **Find y_p:** Use Euler's relation $\sin 2x = \text{Im}(e^{2ix})$. We solve for e^{2ix} and take the imaginary part.

$$y_p = \text{Im}\left[\frac{1}{D^2+4}xe^{2ix}\right]$$

Apply Exponential Shift (Shift e^{2ix} to the front, replace $D \to D + 2i$):

$$y_p = \text{Im}\left[e^{2ix}\frac{1}{(D+2i)^2+4}x\right] = \text{Im}\left[e^{2ix}\frac{1}{D^2+4iD-4+4}x\right]$$

$$y_p = \text{Im}\left[e^{2ix}\frac{1}{4iD(1+\frac{D}{4i})}x\right]$$

Expand the binomial up to D: $(1+\frac{D}{4i})^{-1} \approx 1 - \frac{D}{4i}$

$$y_p = \text{Im}\left[e^{2ix}\frac{1}{4iD}\left(1-\frac{D}{4i}\right)x\right] = \text{Im}\left[e^{2ix}\frac{1}{4iD}\left(x-\frac{1}{4i}\right)\right]$$

The operator $1/D$ means integrate with respect to x:

$$y_p = \text{Im}\left[e^{2ix}\frac{1}{4i}\left(\frac{x^2}{2}-\frac{x}{4i}\right)\right] = \text{Im}\left[e^{2ix}\left(\frac{x^2}{8i}-\frac{x}{16i^2}\right)\right]$$

Multiply the first term by i/i: $\frac{1}{i} = -i$. Since $i^2 = -1$:

$$y_p = \text{Im}\left[(\cos 2x + i\sin 2x)\left(\frac{x}{16} - i\frac{x^2}{8}\right)\right]$$

Expand and collect only imaginary parts (terms with i):

$$\text{Im} = -\frac{x^2}{8}\cos 2x + \frac{x}{16}\sin 2x$$

3. **Final General Solution:**

$$y = C_1\cos 2x + C_2 \sin 2x - \frac{x^2}{8}\cos 2x + \frac{x}{16}\sin 2x$$

Question 25: Solve: $\frac{d^2y}{dx^2} + 4y = e^x + \sin 2x$

Solution:

1. **Find y_c:** $m^2 + 4 = 0 \implies m = \pm 2i$.

$$y_c = C_1 \cos 2x + C_2 \sin 2x$$

2. **Find y_{p1} (Exponential part):**

$$y_{p1} = \frac{1}{D^2+4}e^x = \frac{1}{1^2+4}e^x = \frac{1}{5}e^x$$

3. **Find y_{p2} (Trigonometric part):**

$$y_{p2} = \frac{1}{D^2+4}\sin 2x$$

Sub $D^2 = -4 \implies 0$. Failure. Multiply by x, differentiate denominator:

$$y_{p2} = x\frac{1}{2D}\sin 2x = \frac{x}{2}\int \sin 2x dx = \frac{x}{2}\left(-\frac{\cos 2x}{2}\right) = -\frac{x}{4}\cos 2x$$

4. **Final General Solution:**

$$y = C_1\cos 2x + C_2\sin 2x + \frac{1}{5}e^x - \frac{x}{4}\cos 2x$$